How Long Should a Contact-Induced Response Last?

Statistical Mechanics of Self-Propelled Organisms (Group 3)

Bothe, M., Poliakov, A., Lardet, E., Pruessner, G., Bertrand, T., & Bordeu, I. (2026). *Communications Physics*, 9, Article 204.

Theoretical base: Bothe et al. (2026)

communications physics

Article

A Nature Portfolio journal

https://doi.org/10.1038/s42005-026-02619-2

Contact-induced differential persistence as a mechanism for cell sorting

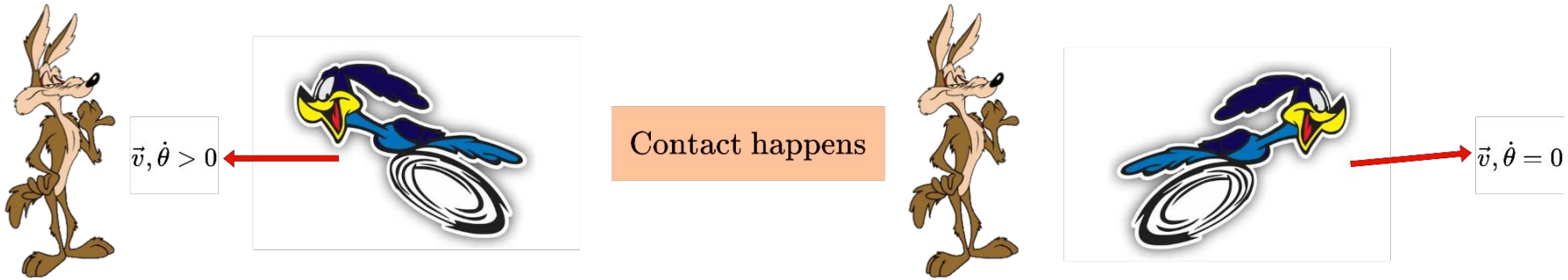
Check for updates

Marius Bothe [1,4], Alexei Poliakov[2,4], Eloise Lardet [1], Gunnar Pruessner[1], Thibault Bertrand [1] & Ignacio Bordeu [3]

Bothe, M., Poliakov, A., Lardet, E., Pruessner, G., Bertrand, T., & Bordeu, I. (2026). *Communications Physics*, 9, Article 204.

Differential persistence

"... temporary increase in the persistence of motion of a cell of one type after interacting with a cell of another type."



Bothe, M., Poliakov, A., Lardet, E., Pruessner, G., Bertrand, T., & Bordeu, I. (2026). *Communications Physics*, 9, Article 204.

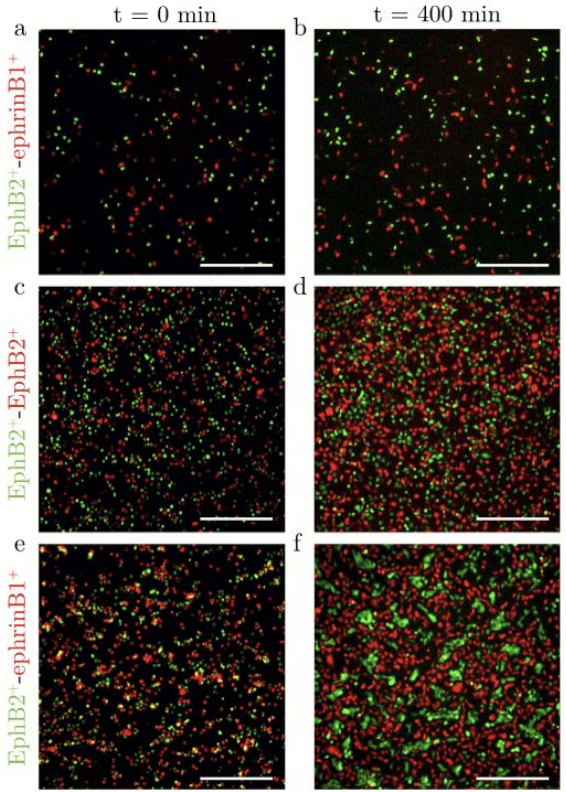
Theoretical base: Bothe et al. (2026)

Does a transient change in the persistence of one cell type activated by cells with a different type drive cell sorting in cell mixtures?

Model and simulations: minimal phenomenological model of two species of interacting active Brownian particles (ABPs)

Differential persistence induces cell sorting

- Differential persistence alone was sufficient to induce cell sorting.
- Green particles formed dense clusters surrounded by dispersed red particles.
- Lower particle densities and greater mobility favored segregation.



Bothe, M., Poliakov, A., Lardet, E., Pruessner, G., Bertrand, T., & Bordeu, I. (2026). *Communications Physics*, 9, Article 204.

Theoretical base: Bothe et al. (2026)

Effect of contact-induced reorientations and differential adhesion:

- Contact-induced reorientation alone promoted segregation but did not produce asymmetric clustering;
- Differences in persistence determined which cell population formed clusters.
- Differential adhesion independently promoted sorting, particularly at higher adhesion strengths.

Temporal dynamics of cell sorting:

- Sorting efficiency varied according to particle density and interaction mechanism;
- Differential persistence combined with reorientation produced the fastest and strongest sorting.

Initial Scenario

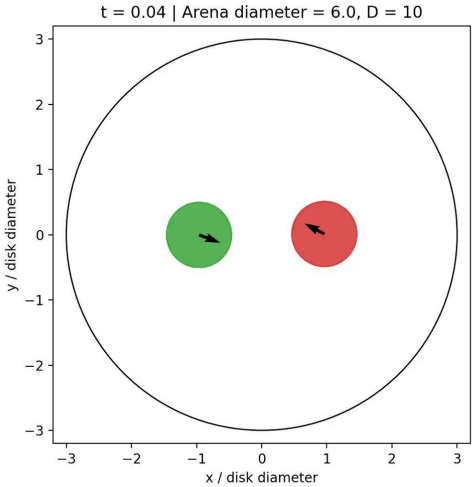
Two interacting cells

Consider a red and a green self-propelled disk of diameter σ and speed v_0 in a circular arena of diameter L. Their positions $\mathbf{r}_i$ and orientations $\mathbf{n}_i = (\cos\theta_i, \sin\theta_i)$ obey

$$\dot{\mathbf{r}}_i = v_0 \mathbf{n}_i + \mu \mathbf{F}_i, \qquad \dot{\theta}_i = \sqrt{2D_i}\,\eta_i(t), \tag{1}$$

Here $\mathbf{F}_i$ is the repulsive force, μ the mobility, and η_i Gaussian white noise. Neglect translational diffusion. Follow Ref. [1] for disk–disk repulsion and reorientation upon contact. Use similar elastic repulsion at the wall, beginning when the disk edge touches it.

Two-cell Interaction Model



Repulsive force

$$\mathbf{F}_i = k\delta\hat{\mathbf{n}} \tag{2}$$

$$\delta = \begin{cases} \sigma - r_{ij}, & \text{disk-disk interaction} \\ \|\mathbf{r}_i\| - \left(\frac{L}{2} - \frac{\sigma}{2}\right) & \text{disk-arena interaction} \end{cases} \tag{3}$$

where $\hat{\mathbf{n}}_{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}$, $\hat{\mathbf{n}}_{\text{arena}} = -\frac{\mathbf{r}_i}{||\mathbf{r}_i||}$, k = repulsion constant, σ = disk diameter and L = arena diameter.

For the simulation we used: speed of the self-propelled disk (v_0) = 1.0, mobility (μ)= 1.0, L = 6.0, σ = 1.0, angular diffusion (D) = 10.0, k = 50.0

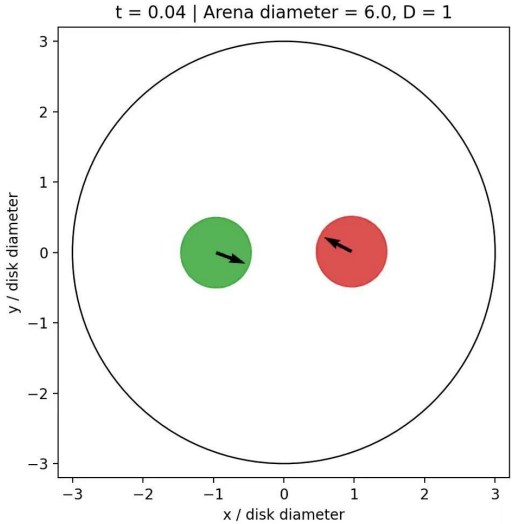
Introducing a disturbance

A temporary change in persistence

Both disks normally have rotational diffusion coefficient D. Upon contact, they reorient as in Ref. [1]. The *green* disk also reduces its rotational diffusion to $D^+ < D$ for a duration τ_a, then returns to D. The red disk retains D. A new response begins at each new encounter, not at every step of a continuous contact.

Vary τ_a with D^+ fixed. It is the *response duration*, not the orientational persistence time $1/D^+$; Ref. [1] calls it τ_p.

Introducing a disturbance



For the green disk:

$$D_{\text{rot}} = \begin{cases} D^{+} = 0.0, & \text{if } t_{\text{contact}} \leq \tau_a \\ D = 10.0, & \text{normal state} \end{cases} \tag{4}$$

But the red disk maintains its behaviour:

$$D_{\text{rot}} = D = 10.0 \quad (\text{constant}) \tag{5}$$

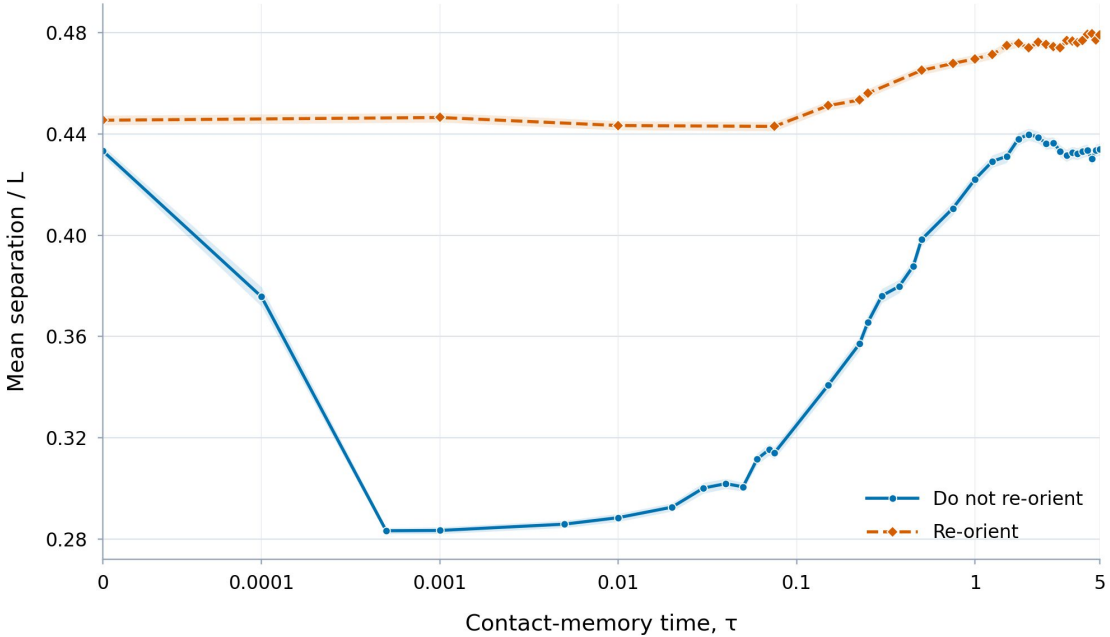
How far apart they remain?

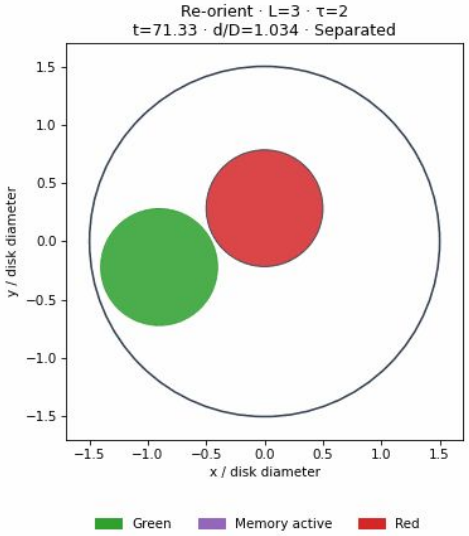
Measure the steady-state distribution $P(s)$ of the separation $s = |\mathbf{r}_{\text{green}} - \mathbf{r}_{\text{red}}|$ and its mean $\langle s \rangle$ for several τ_a. Compare with $\tau_a = 0$, which retains reorientation but removes the persistence change. Do longer responses keep the disks farther apart?

With re-orientation: **YES!**

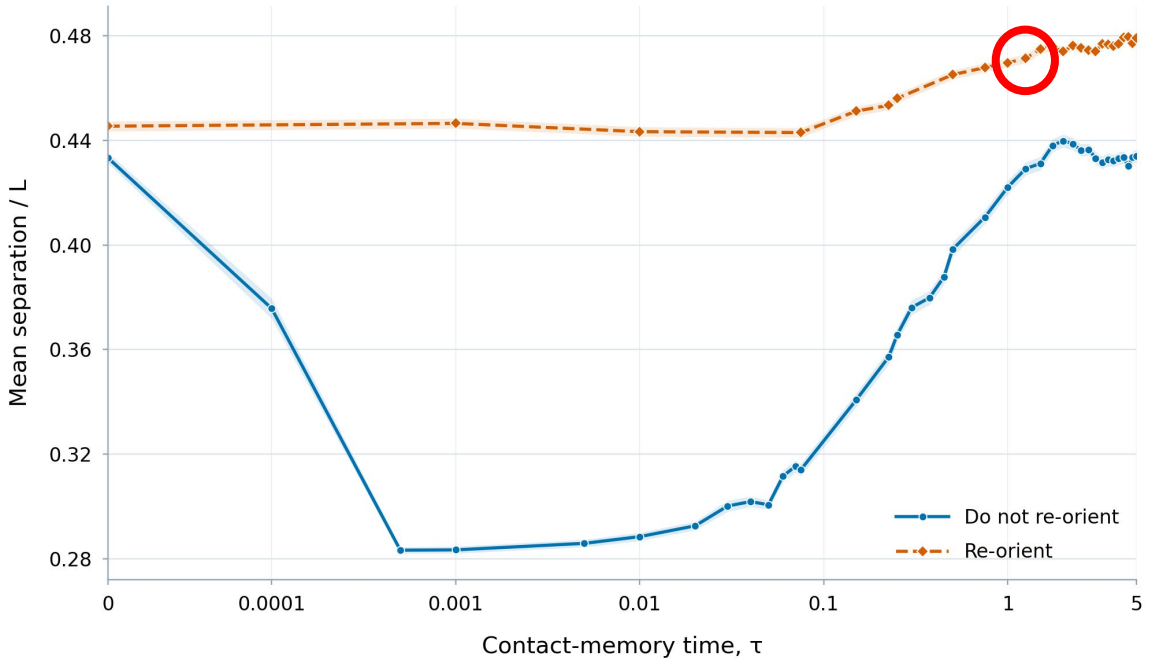
Without re-orientation… not really, here's how:

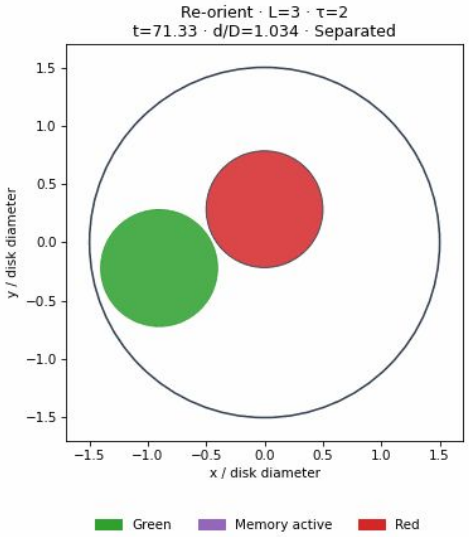
Is there a preferred response duration?

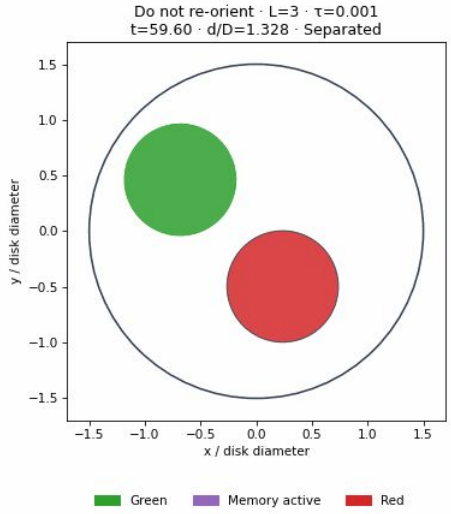




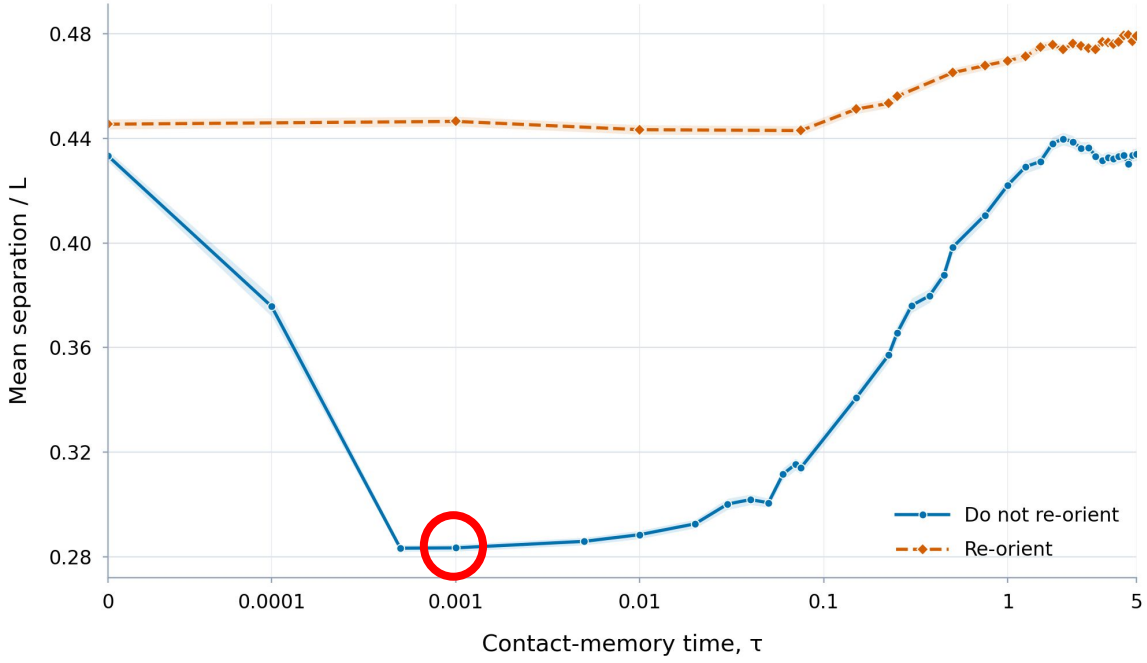
Is there a preferred response duration?







Is there a preferred response duration?



Conclusion & Next Steps

What are the differences from our project and Bothes et al?

Change in optimal?

What happens with a lower density and how does that scale?

What were the limitations?



Thank you! Beep Beep!